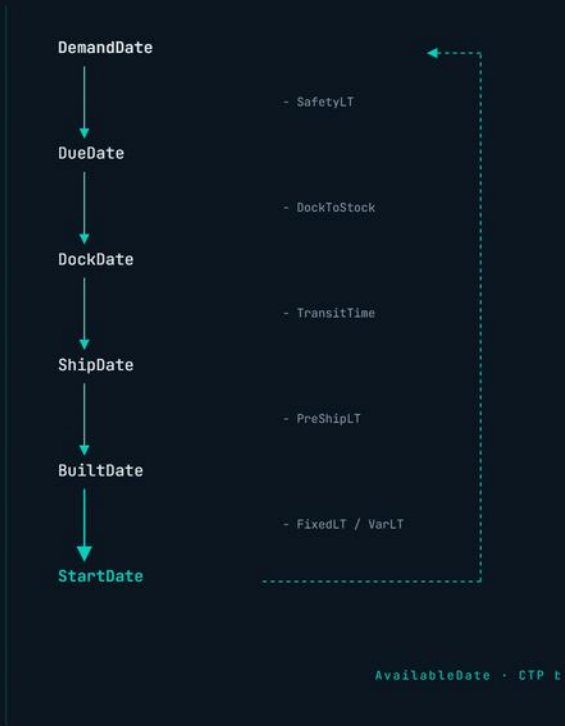

Date Chain *Autopsy.*

A 10-slide decomposition of how Maestro propagates a single date through the supply graph — and the one branch *it never imports*.

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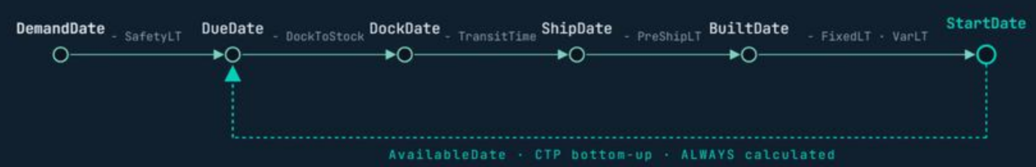
Seven links, one hidden branch. *Six are imported. One is not.*

Maestro stores eight dates per supply order. Five are produced by a top-down subtraction chain that begins at `DemandDate` and ends at `StartDate`. The eighth — `AvailableDate` — is the only date Maestro *always* calculates and never imports, propagated bottom-up by CTP through the BOM. Confusing the two directions is the single most common cause of a date-chain ticket.



TOP-DOWN

5 subtractions, 6 nodes, all imported from `ScheduledReceipt` when LTO=Yes.

BOTTOM-UP

`AvailableDate` derives from CTP propagation through BOM components — never read from the order.

01/02

DEMANDDATE
DUEDATE
DOCKDATE
SHIPDATE
BUILTDATE
STARTDATE

The first cut *is the safety lead time.*

Maestro starts at the demand horizon and walks backwards. The first subtraction is the safety lead time — a buffer that pulls the receipt date earlier than the customer needs the part.

```
DueDate = DemandDate - SafetyLT
```

`DemandDate`	When the customer needs goods on the floor — the chain anchor.	IMPORTED
`SafetyLT`	Calendar-day buffer pulled from `PartSource` or `Part`.	IMPORTED
`DueDate`	When the order must be receipted in stores — the first internal date.	COMPUTED

FIELD-TESTED GOTCHA

When `LTO = Yes`, every link below `DueDate` is read from `ScheduledReceipt` wholesale. You cannot override only one segment — it is all or nothing.

The dock-to-stock interlude — *where transit hides.*

Two short subtractions sit between the warehouse acceptance date and the moment the carrier leaves the source plant. They look identical on a Gantt; they cost very different things in real life.

03 `DockDate = DueDate - DockToStock`

Inbound put-away buffer — internal time to inspect, label, and shelve. **Quality-managed parts inflate this silently.**

04 `ShipDate = DockDate - TransitTime`

Carrier window from supplier dock to your dock. On STO, **this is the field that disappears under LTO=Yes.**

> WHY THIS MATTERS
>
> A 14-day `DockToStock` on a make-to-stock API and a 14-day `TransitTime` on its finished good produce identical Gantts. They are not the same buffer. One you can compress with a quality waiver. The other you cannot compress without changing carrier.

Three more subtractions *to the floor.* *Then the chain stops.*

05

PRE-SHIP

`BuiltDate · ShipDate - PreShipLT`

`BuiltDate = ShipDate - PreShipLT`

Wash, label, palletize, stage at the outbound dock. Often configured as zero on commodity goods **and as 5–7 days on regulated material** (pharma batch release, hazmat documentation).

06

PRODUCTION

`StartDate · BuiltDate - FixedLT - VarLT(qty)`

`StartDate = BuiltDate - FixedLT - VarLT(qty)`

The release date for production. **VarLT scales with order quantity** — increasing the lot size pushes `StartDate` earlier, not `BuiltDate` later.

07

RELEASE (STO ONLY)

`Transfer orders · demand peg = BuiltDate`

`Source-side demand peg → BuiltDate (`*not* `StartDate)`

One subtle Ch.14 footnote planners miss: a transfer order pegs its source plant on the **BuiltDate**, not the `StartDate`. Misreading this turns into ghost shortages on the sending site.

REMINDER The chain ends at `StartDate`. Anything earlier — material availability, capacity windows — is a different question, with a different engine, on the next slide.

Six dates flow *down.*
One date flows *up.*

TOP-DOWN · IMPORTED

6 dates

All read from `ScheduledReceipt` when `LTO=Yes`; otherwise re-derived from `PartSource` lead-time fields. The engine only computes them — it does not invent them.

- `DemandDate`
- `DueDate`
- `DockDate`
- `ShipDate`
- `BuiltDate`
- `StartDate`

BOTTOM-UP · NEVER IMPORTED

1 date

`AvailableDate` is the earliest date the order *could* be ready given the availability of every component below it in the BOM. CTP propagates it bottom-up. No SAP record ever sets it. No `.tab` file can carry it.

- `AvailableDate`

MENTAL MODEL

"The top-down chain says when this order should arrive.
The bottom-up branch says when it actually can. Their disagreement is the planner's daily inbox."

SOURCE: MAESTRO DATA MODEL & ALGORITHM GUIDE · CH. 14, CH. 19 (CTP)

Three values, *three different machines.*

`SupplyStatus.LTO` is rendered as a single column in the planning sheet, but it switches Maestro between three completely different lead-time engines. The names suggest a sliding scale; the behaviours are discrete states.

MODE A

No

`DueDate = Today + FixedLT + DockToStock`

SOURCE	`PartSource` · all LT fields
ENGINE	Recompute on every plan
USE	Planned orders, requisitions, internal MTO

Default for forward-planning. Lead times are engineering knobs and the engine respects them.

MODE B · DESTRUCTIVE

Yes

`All dates ← ScheduledReceipt (no recalc)`

SOURCE	`ScheduledReceipt` as imported
ENGINE	No date recompute, all or nothing
USE	Firm POs, work orders honoured by SAP

Trap on transfers (STO): the engine drops transit time entirely if the source plant is not feeding it.

MODE C

FromOrder

`DueDate = LT field on the order itself`

SOURCE	`Order.LT` column
ENGINE	Per-order override, no master data
USE	`OpenRecommendSTO`, supplier-confirmed slots

Reserved for STO recommendations where each order carries its own committed lead time from the carrier.

ONE MORE LEVER `IgnoreStartDate=Yes` overrides the imported `StartDate` and forces Maestro to re-derive it from the chain. `IgnoreCalcDate=Yes` freezes the `DueDate` against any recompute. They are independent — and they layer.

EXHIBIT A

STO RECOMMENDATION · PLANT 8359 → 8712 · LTO=YES

When LTO=Yes *on a transfer order,* Maestro silently drops your transit time.

SYMPTOM

Planner flags an `OpenRecommendSTO` with a `StartDate` matching the requested ship date — no transit window at all.

NAIVE READ

The first hypothesis is always the carrier calendar. It is never the carrier calendar.

WALK THE CHAIN

Top-down chain says `ShipDate = DockDate - TransitTime`. `TransitTime` on `PartSource` is correct. It is not being read.

EVIDENCE

`SupplyStatus.LTO = Yes` on the recommendation. All eight dates are read from `ScheduledReceipt` wholesale — and the inbound feed never carried `TransitTime`.

RULE

You cannot override *one* link of the chain when `LTO=Yes`. It is all or nothing.

EXHIBIT B · FIELD-BY-FIELD

The transit time is on PartSource. Maestro is reading ScheduledReceipt.

FIELD	PARTSOURCE	USED BY ENGINE
FixedLT	14 d	– ignored
TransitTime	7 d	– ignored
DockToStock	2 d	– ignored
All dates	–	ScheduledReceipt

RESOLUTION

For STO recommendations: set `LTO = FromOrder`, set `IgnoreStartDate = Ignore`, and let the carrier window from the order itself populate the chain. For confirmed open POs / WOs / WO allocations, `LTO = Yes` remains correct — they should honour SAP's dates verbatim.

CalcStartDate is *not StartDate.*

When Maestro flags an order to reschedule, it does not simply push the chain. It computes a **delta** between the effective due date and the rescheduled due date, then **adds that delta to the existing StartDate.** The order keeps its production-side phasing.

- → Only when **StartDate** is already populated.
- → Only on rescheduled (not net-new) orders.
- → Independent of **LTO**.

RESCHEDULE FORMULA · CH. 14

```
CalcStartDate
= StartDate
+ ( ReschedDate -
EffDueDate )
```

`StartDate`	**The production release date as currently held on the order.** Untouched by the recompute.
`ReschedDate`	**The new target date the engine wants to honour.** Often equals the recomputed DueDate.
`EffDueDate`	**The effective due date the order was last anchored to.** Read from the order, not recomputed.
Δ	**The shift, signed.** Positive = push later, negative = pull in.

WHY IT'S BUILT THIS WAY

The chain protects production phasing. Recomputing StartDate from a fresh DueDate would erase any manual finite scheduling already applied — the delta keeps it intact.

SOURCE: CH. 14

Six dates down. One date *up.*

SOURCES

Maestro Data Model & Algorithm Guide
SU2603 · Ch. 14

Maestro Data Model & Algorithm Guide
SU2603 · Ch. 19 (CTP)

Hypercare casework
2025–2026

i. The chain is a top-down subtraction. `AvailableDate` is the only date Maestro never imports.

ii. `LTO=Yes` reads *every* date from `ScheduledReceipt`. You cannot override one segment.

iii. Transfers peg the source plant on `BuiltDate` — not `StartDate`.

iv. On reschedule, `CalcStartDate = StartDate + (ReschedDate - EffDueDate)`. Production phasing is preserved.

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